

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed a vision framework for new **TI SoC devices**, integrating with the **Linux Yocto build process** to customize the kernel and file system, enabling rapid prototyping for automotive mobility solutions.
- Managed the setup of project repositories according to **Valeo standards** and adapted the toolchain, ensuring consistent development environments for embedded software projects and reducing onboarding time by **15%**.
- Engineered end-to-end **AI solutions** for requirement analysis and code generation, facilitating platform deployment into customer projects and accelerating development cycles by an estimated **20%**.
- Contributed to the development and maintenance of module requirements, design specifications, and module test plans, adhering to **MISRA coding guidelines** and ensuring code quality for embedded systems.
- Executed comprehensive testing strategies, including **black box, integration, and unit testing**, identifying and resolving critical defects to improve platform stability and meet stringent QA requirements.
- Organized and participated in formal code and document reviews, performing **static code analysis** using tools like Klocwork to ensure adherence to coding guidelines and maintain high software quality standards, achieving a **95% reduction** in critical code review findings.

SKILLS

Embedded Systems – Low level software, Firmware, Drivers, Middleware, Hardware-Software Interface (GPIOs, SPI, I2C, UART, Ethernet, Memory)

Development Tools – Linux Yocto, Kernel Customization, File System Customization, Containerization, CMake, Compiler, Linker, Klocwork, Code Beamer

Programming Languages – C, C++, Python

Software Development Lifecycle – Module Requirements, Module Design, Module Test, Black Box Testing, Integration Testing, Unit Testing, SRS, SW Arch, Agile/Scrum, CI/CD

Quality & Standards – MISRA C/C++, ASPICE, ISO15504, QA Requirements

Cloud & Automation – AWS (EC2/S3/Lambda/CloudWatch), Docker, Python Scripting, Jenkins

Data & Analytics – Power BI, DAX, Tableau, R, pandas, NumPy, scikit-learn

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Automotive Vision Framework for Embedded AI (HCL Technologies)

- Developed a novel vision framework for a new **TI SoC device**, enabling advanced AI capabilities for automotive applications, delivered **2 months ahead of schedule**.

- Supported the **Linux Yocto build process**, including kernel and file system customization, and established **container-based build environments** to streamline development and ensure reproducible builds.
- Integrated hardware interfaces such as **SPI and I2C** to port the vision framework to new SoC devices, demonstrating platform functionality on **ECU hardware** and validating performance metrics.

Real-Time Transaction Monitoring Dashboard (HCL Technologies)

- Built and maintained a **payments operations dashboard** for a financial-services client, providing real-time transaction flow monitoring and operational alerting for failed messages.
- Implemented automated data pipelines using **Python and SQL** to ingest and process high-volume transaction data, ensuring data integrity for reconciliation workflows and audit trails.
- Managed production support and incident response for the dashboard in a **GDPR-aware environment**, consistently meeting **99.9% SLA uptime** through proactive monitoring with Splunk and Dynatrace.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)